



YENEPOYA INSTITUTE OF TECHNOLOGY

NH-13, THODAR, MOODBIDRI, DAKSHINA KANNADA-574225

2025-26



**Department of Computer Science and Engineering
(IoT, Cybersecurity including Blockchain technology)**

**Mini Project
Presentation on**

“Iot Based Smart Waste Management System”

PRESENTED BY :

ISRA ZAINAB	(4DM23IC024)
MADEEHA MOHAMMED MUBEEN	(4DM23IC029)
SANIYA NAZ	(4DM23IC050)
SHAMMA SHIRIN MA	(4DM23IC052)

**UNDER THE GUIDANCE OF
Prof. K .AKSHATHA DANGE
Assistant Professor
Dept. of CSE(IoT)**

DATE OF PRESENTATION: 02/12/2025

TABLE OF CONTENT

- Abstract
- Introduction
- Problem Statement
- Objectives
- Methodology
- Contribution
- System Requirements
- Code Snippet
- Hardware Prototype
- Wokwi Stimulation Prototype
- Serial monitor output
- Results
- Challenges faced
- Future scope
- Conclusion
- References

ABSTRACT

- The current waste collection system is inefficient due to random truck schedules and overflowing bins
- Smart bins with sensors can detect fill levels and send real-time alerts
- These bins connect to a central monitoring system via the internet.
- The system assigns the nearest garbage truck to collect waste when a bin is full.
- This optimizes routes, saves time, prevents overflow, and reduces traffic issues.

INTRODUCTION

- Sensors in smart bins check how full they are and send this information to a central system.
- The central system helps plan when and where to collect trash to stop bins from overflowing.
- Smart systems use smart programs to find the best routes for garbage trucks to save fuel and time.
- They make sure all areas of the city get fair and timely garbage collection.
- Challenges include making the system work for large cities, keeping data safe, and getting people to use it well

PROBLEM STATEMENT

- Cities make a lot of waste, but old systems can't handle it well.
- Bins overflow, causing health risks, pollution, and traffic problems.
- Garbage trucks follow fixed paths, wasting time and resources.
- Systems don't know bin status in real time to fix problems quickly.
- Better systems with smart sensors can help manage waste faster and cleaner.

OBJECTIVES

- Manage growing city waste effectively with better systems.
- Stop bins from overflowing to reduce health and pollution risks.
- Make garbage collection faster and save time by avoiding fixed routes.
- Know bin status in real time to act quickly and prevent problems.
- Use smart sensors and technology to keep cities cleaner and waste management smart.

METHODOLOGY

- Put sensors inside each bin to check how full it is.
- Send the bin level data to an online/cloud system.
- Show all bins and their levels on a dashboard or app.
- Set warning and full limits (for example 70% and 90%).
- When a bin is full, create a pickup request automatically.
- Send the bin location and route to the nearest garbage truck.
- After pickup, update the bin level and close the request or send a new alert if still full.

CONTRIBUTION BY EACH MEMBER

ISRA – Simulation & Serial

- TASK 1: Wokwi Simulation
 - Complete circuit (ESP32+sensors+LCD)
 - Fill simulation (1-10% random steps)
 - Full cycle testing
- TASK 2: Serial Monitor & Debug
 - Real-time fill% logging
 - WiFi status monitoring
 - Error handling code

MADEEHA –Blynk & Cloud

- TASK 1: Blynk Dashboard
 - Setup V1(distance), V2(fill%), V5(map)
 - Create alert_75, alert_95 events
 - Mobile app testing + screenshots
- TASK 2: Location Tracking
 - Google Maps integration
 - GPS coordinates (12.9716, 77.5946)
 - Multiple bin support (future)

SANIYA – Firmware & Testing

- TASK 1: Alert Logic Programming
 - 75% alert → Truck 1 only
 - 95% alert → Both trucks
 - Overflow detection (100%)
 - Collection simulation (8s timer)
- TASK 2: Hardware Testing
 - Real bin fill tests (50/75/95/100%)
 - WiFi stability testing
 - Alert timing verification

SHIRIN– Sensors & Display

- TASK 1: Ultrasonic Sensor System
 - Wire HC-SR04 (trigPin1=13, echoPin1=14)
 - Calibrate 40cm bin depth → fill%
 - Test real distances (0-40cm)
- TASK 2: LCD Display Integration
 - I2C LCD wiring (0x27, 20x4)
 - Status messages (OK/Moderate/Critical/OVERFLOW)
 - Real-time fill% display testing

SYSTEM REQUIREMENTS

Hardware requirements:

- ESP8266 (WiFi microcontroller)
- Ultrasonic Sensor (HC-SR04)
- Neo 6M-GPS Module
- LCD with i2c (16X2)-Display
- Power supply
- SIM900A-GSM Module

Software requirements:

- Arduino IDE
- C programming
- Serial Monitor

CODE SNIPPETS

SENSOR / LEVEL READING:

```
// Read ultrasonic distance
float levelDistance = measureDistance(trigPin1, echoPin1);
// Convert to simulated fill percentage (0–100 %)
if (!waitingForCollection && simFill < 100 &&
    millis() - lastFillUpdate >= holdDuration) {
    lastFillUpdate = millis();
    int step = random(1, 11); // +1 to +10 %
    simFill += step;
    if (simFill > 100) simFill = 100;
    holdDuration = random(1500, 4000); // 1.5–4 s
}
```

WIFI CONNECTION:

```
WiFi.begin(ssid, pass);
while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}
Serial.println(" Connected!");
```

GARBAGE COLLECTION & SERIAL OUTPUT:

```
// Start truck timer after 95 % alert
if (alert95Sent && !waitingForCollection) {
    waitingForCollection = true;
    collectionStartTime = millis();
}

// After 8 s, empty the bin
if (waitingForCollection &&
    millis() - collectionStartTime > 8000) {
    simulateCollection(); // simFill -> 0, flags reset
}

// Serial print: fill level in cm and %
const float BIN_DEPTH_CM = 40.0;
float fillLevelCm = (simFill / 100.0) * BIN_DEPTH_CM;

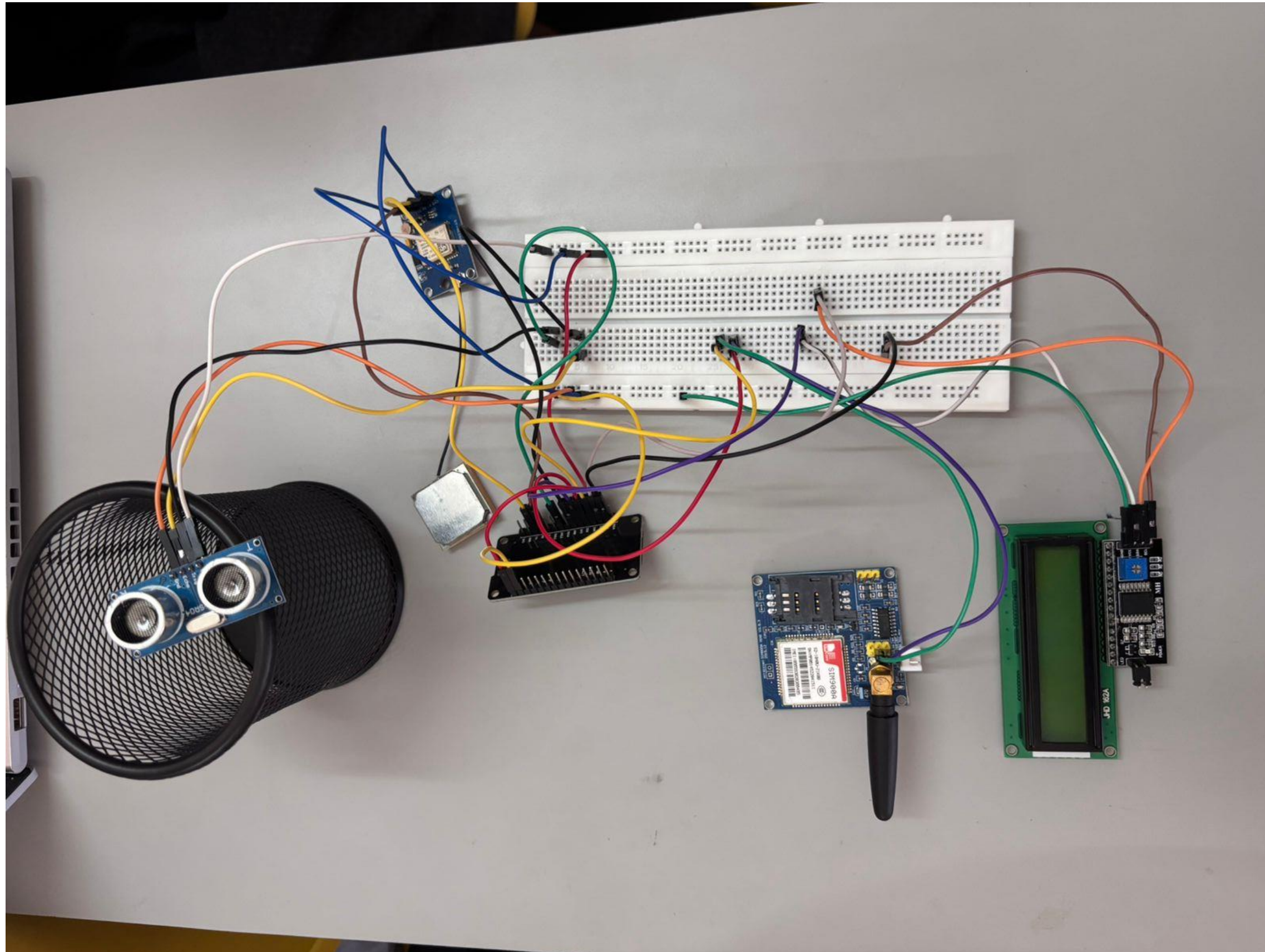
Serial.print("Fill level: ");
Serial.print(fillLevelCm, 1);
Serial.print(" cm | Fill: ");
Serial.print(simFill);
Serial.println(" %");
```

OVERFLOW & ALERT DETECTION:

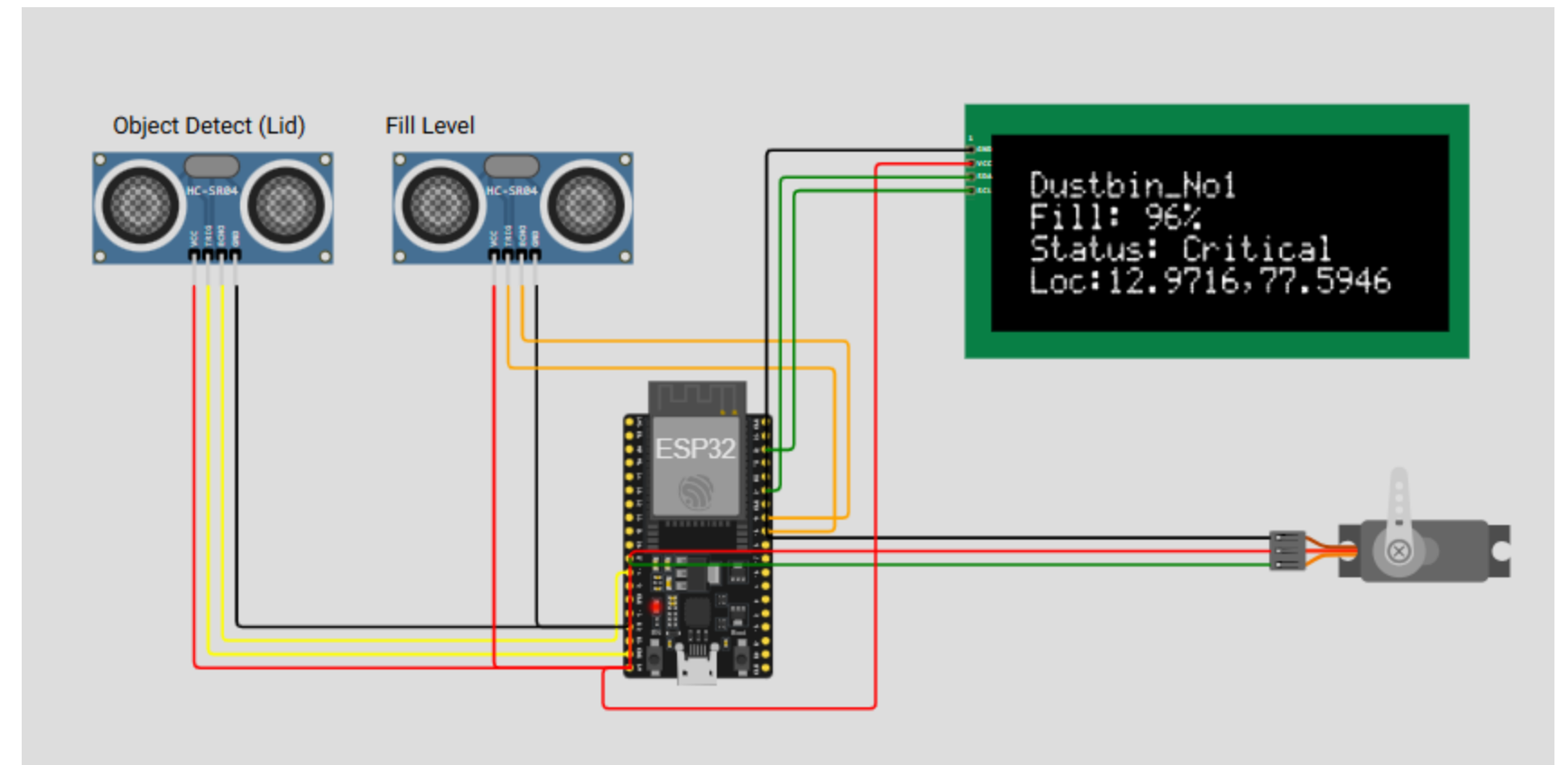
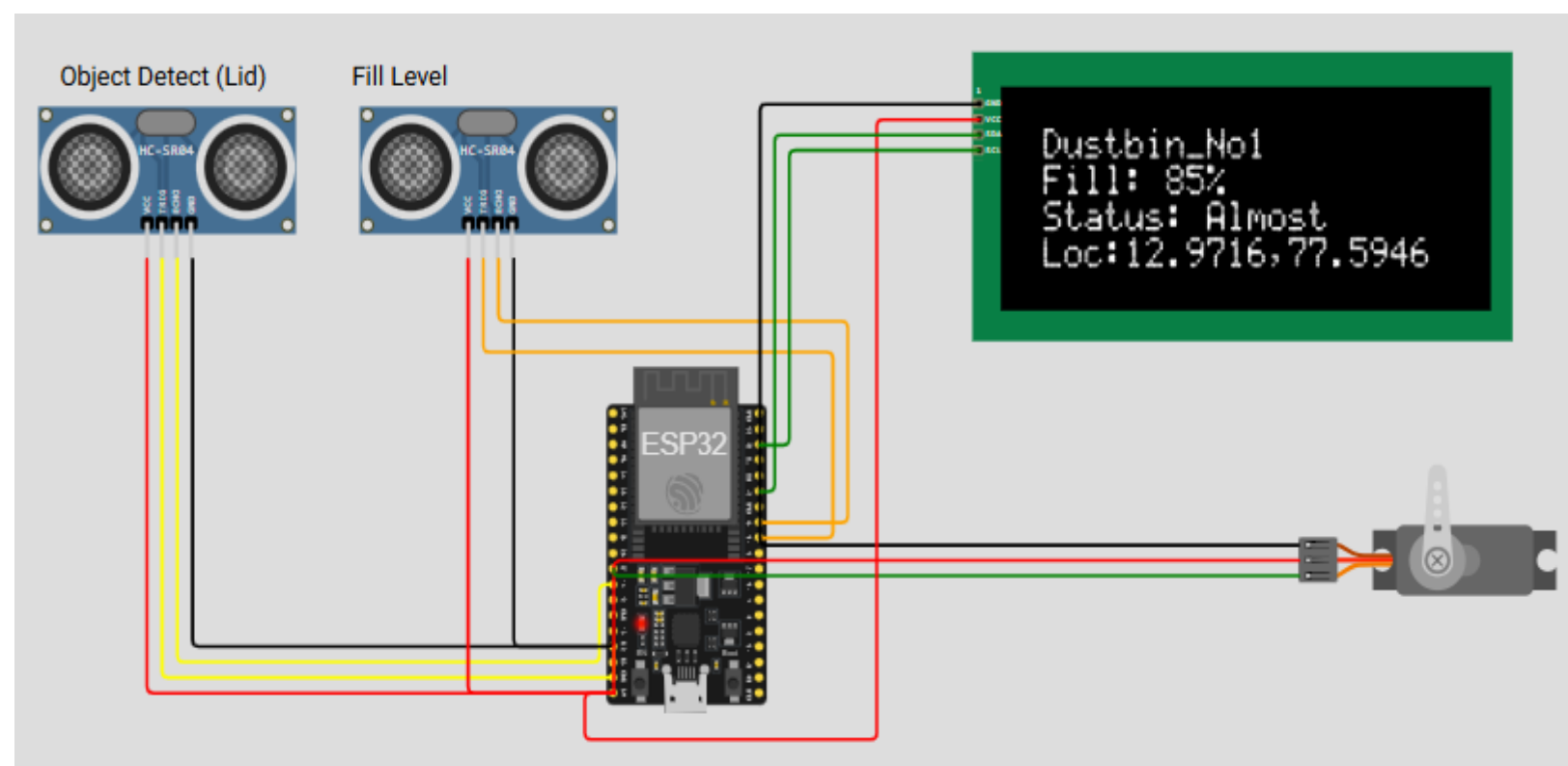
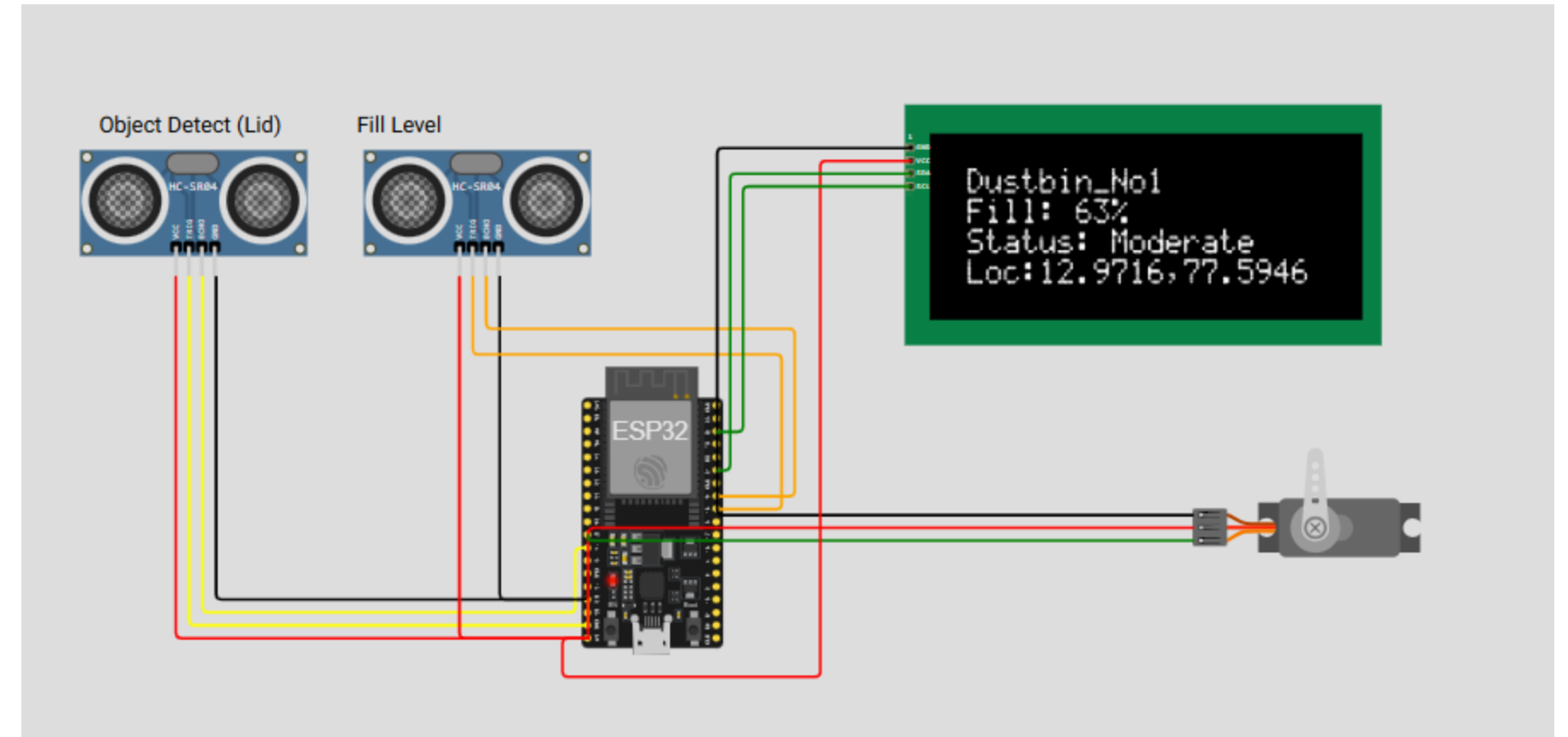
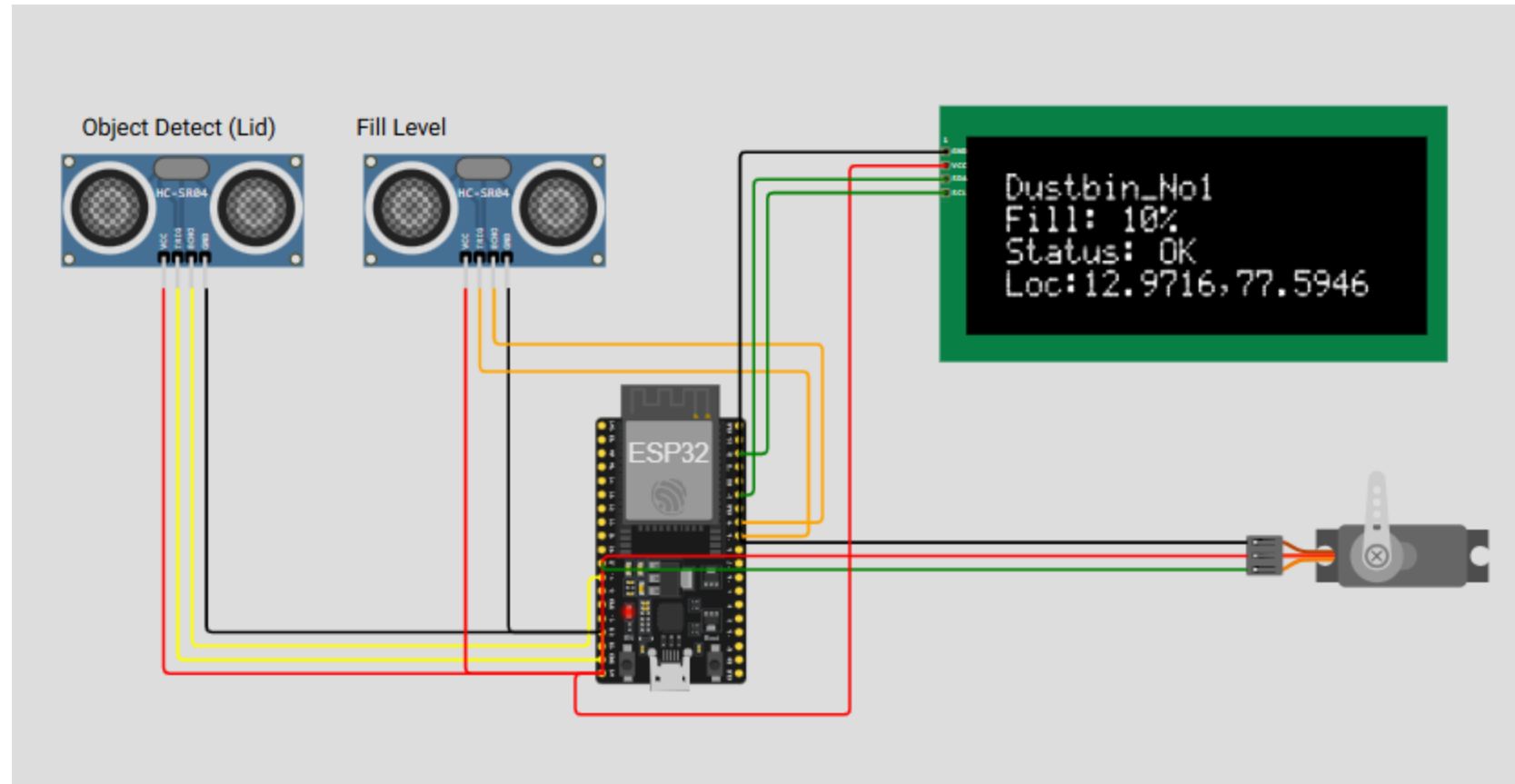
```
// Overflow at 100 %
if (simFill == 100 && !overflowPrinted) {
    Serial.println("***** WARNING: BIN OVERFLOW
*****");
    overflowPrinted = true;
}

// Alerts at 75 % and 95 %
if (simFill >= 75 && !alert75Sent) {
    sendAlert75(simFill);
    alert75Sent = true;
}
if (simFill >= 95 && !alert95Sent) {
    sendAlert95(simFill);
    alert95Sent = true;
}
```

HARDWARE PROTOTYPE



WOKWI SIMULATION PROTOTYPE



SERIAL MONITOR OUTPUT

```
Fill level: 30.0 cm | Fill: 75 %

--- ALERT: 75% LEVEL REACHED ---
Truck 1 is notified to collect this bin.
Fill: 75 %
Maps: https://maps.google.com/?q=12.971600,77.594600
-----

Fill level: 30.0 cm | Fill: 75 %
Fill level: 30.0 cm | Fill: 75 %
Fill level: 30.0 cm | Fill: 75 %
Fill level: 30.0 cm | Fill: 75 %
Fill level: 30.4 cm | Fill: 76 %
Fill level: 30.4 cm | Fill: 76 %
Fill level: 30.4 cm | Fill: 76 %
Fill level: 30.4 cm | Fill: 76 %
Fill level: 32.0 cm | Fill: 80 %
Fill level: 32.0 cm | Fill: 80 %
Fill level: 32.0 cm | Fill: 80 %
Fill level: 32.0 cm | Fill: 80 %
Fill level: 32.0 cm | Fill: 80 %
Fill level: 33.2 cm | Fill: 83 %
Fill level: 33.2 cm | Fill: 83 %
Fill level: 33.2 cm | Fill: 83 %
Fill level: 35.6 cm | Fill: 89 %
Fill level: 35.6 cm | Fill: 89 %
Fill level: 35.6 cm | Fill: 89 %
```

```
=== FINAL ALERT: 95% LEVEL ===
All available garbage collectors (Truck 1 & 2) notified.
Fill: 98 %
Maps: https://maps.google.com/?q=12.971600,77.594600
=====

Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %

##### GARBAGE COLLECTED #####
Garbage truck has emptied the bin.
Bin is now EMPTY (0%). Filling cycle restarts.
#####
```

RESULTS

- Bin level sensors were successfully integrated with the controller (ESP).
- Real-time fill level data was obtained from all bins.
- Data was sent to the cloud platform without errors.
- Automatic pickup alerts and optimized routes worked as expected.
- Bin overflow incidents were reduced during testing.
- A fully working smart bin hardware prototype was demonstrated.

CHALLENGES FACED

- Difficulty connecting the bin's controller to Wi-Fi and keeping the connection stable.
- Ultrasonic/IR sensor sometimes gave shaky or wrong level readings.
- Setting correct fill limits (empty, warning, full) needed several trials.
- Cloud data and alerts from this bin were sometimes delayed or missed.
- Power supply and wiring inside the single bin had to be safely and neatly managed.

FUTURE SCOPE

- Add AI to predict waste levels and optimize collection.
- Use solar power for bins to save energy.
- Include waste sorting robots for better recycling.
- Expand to cover larger city areas easily.
- Improve mobile apps for real-time monitoring and alerts.
- Integrate with other smart city services for efficiency.

CONCLUSION

- A working smart waste management system was successfully developed.
- Real-time monitoring of bin fill level was achieved using IoT.
- Cloud platform enabled remote viewing of bin status and alerts.
- System can trigger timely collection and avoid bin overflow.
- Helps reduce collection trips, saves resources, and keeps the area cleaner.

REFERENCES

- A. J. Fathima, R. Raman, S. Omkumar and J. Omana, "IoT-Based Intelligent System for Garbage Level Monitoring in Smart Cities," 2023 International Conference on IoT, Communication and Automation Technology (ICICAT), Gorakhpur, India, 2023, pp. 1-5, doi: 10.1109/ICICAT57735.2023.10263763.
- T. M. N. Vamsi, G. Kalyan Chakravarthi, P. Lanka and B. Divakar, "An IoT Based Smart Garbage Monitoring and Disposal Support System," 2021 5th International Conference on Computing Methodologies and Communication (ICCMC), Erode, India, 2021, pp. 438-442, doi:10.1109/ICCMC51019.2021.9418289.
- I. Sosunova and J. Porras, "IoT-Enabled Smart Waste Management Systems for Smart Cities: A Systematic Review," in IEEE Access, vol. 10, pp. 73326-73363, 2022, doi: 10.1109/ACCESS.2022.3188308.
- M. A. Rahman, S. W. Tan, A. Taufiq Asyhari, I. F. Kurniawan, M. J. F. Alenazi and M. Uddin, "IoT-Enabled Intelligent Garbage Management System for Smart City: A Fairness Perspective," in IEEE Access, vol. 12, pp. 82693-82705, 2024, doi: 10.1109/ACCESS.2024.3412098.
- M. Ghahramani, M. Zhou, A. Molter and F. Pilla, "IoT-Based Route Recommendation for an Intelligent Waste Management System," in IEEE Internet of Things Journal, vol. 9, no. 14, pp. 11883-11892, 15 July 2022, doi: 10.1109/JIOT.2021.3132126.

THANK YOU